

Practical_6(b)

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if (fork() == 0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);
    }

    if (fork() == 0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);

        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

Output

```
ketana_sindhu@KetaZone:~$ nano PipeGrip.c
ketana_sindhu@KetaZone:~$ gcc PipeGrip.c -o PG
ketana_sindhu@KetaZone:~$ ./PG
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 501 Sep 18 05:37 AddressTranslation.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 345 Sep 18 03:54 DemandPage.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 110 Jul 31 09:39 DisplayProcessID.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 220 Sep 29 05:21 Fifo.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 254 Sep 29 05:22 FifoRead.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 292 Sep 29 05:26 Fifo_Write.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 264 Jul 31 09:48 File.c
-rw-r--r-- 2 ketana_sindhu ketana_sindhu 77 Aug 4 14:54 FirstFile.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 104 Aug 24 05:09 Fork.c
-rw-r--r-- 2 ketana_sindhu ketana_sindhu 77 Aug 4 14:54 HELLO.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 0 Aug 4 14:52 KL.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 233 Sep 18 05:40 MemoryAllocate.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 287 Jul 31 09:44 PID.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 230 Aug 24 05:22 ParentAndChild.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 505 Sep 29 05:36 PipeGrip.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 402 Sep 29 04:13 Practical3.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 369 Sep 29 04:13 Practical4.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 158 Jul 31 09:36 ReadUser.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 1274 Sep 29 04:52 RouundRobin.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 141 Sep 18 04:08 Segmentation.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 528 Sep 29 05:10 SigChild.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 551 Sep 29 05:12 SigPipe.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 340 Sep 29 05:05 SigTerm.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 420 Sep 29 05:15 Sigsegv.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 0 Aug 4 14:51 Sindhu.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 0 Aug 24 05:53 Skill1.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 413 Sep 18 03:46 Variable.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 136 Sep 18 03:47 Virtual.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 0 Sep 18 03:10 Virtual.c.save
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 274 Jul 31 09:46 Wait.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 455 Aug 7 09:52 WeekTwo.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 464 Aug 7 10:07 WeekTwoONE.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16032 Sep 28 15:03 cow
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 293 Sep 28 15:03 cow.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16096 Sep 29 04:43 fcfs
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 692 Sep 29 04:43 fcfs.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16312 Aug 7 09:53 file.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16040 Sep 6 06:50 fork.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 180 Sep 6 06:43 fork_demo.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16136 Sep 18 05:40 malloc
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 531 Sep 28 14:38 memory.c
-rwxr-xr-x 1 ketana_sindhu ketana_sindhu 16088 Sep 28 14:55 memory_alloc
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 237 Sep 28 14:55 memory_alloc.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 289 Sep 6 12:49 pipe.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 0 Sep 6 12:46 pipe.c.save
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 1211 Sep 29 04:50 priority.c
lrwxrwxrwx 1 ketana_sindhu ketana_sindhu 9 Aug 4 15:47 shortcut -> FirstFile
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 311 Sep 29 04:59 signals.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 967 Sep 29 04:48 sjfs.c
-rw-r--r-- 1 ketana_sindhu ketana_sindhu 159 Aug 24 05:20 wait.c
ketana_sindhu@KetaZone:~$ |
```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT Received\n");

    if (sig == SIGTERM)
        printf("\nSIGTERM Received\n");

    if (sig == SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    while (1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}|
```

Output

```
ketana_sindhu@KetaZone:~$ nano Signals.c
ketana_sindhu@KetaZone:~$ gcc Signals.c -o SG
ketana_sindhu@KetaZone:~$ ./SK
-bash: ./SK: No such file or directory
ketana_sindhu@KetaZone:~$ ./SG
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
```

```
ketana_sindhu@KetaZone:~$ ps
  PID TTY          TIME CMD
 1912 pts/2        00:00:00 bash
 1948 pts/2        00:00:00 ps
```